

Black and Scholes Solution

J.M Zarate

Department of Mathematics and Physics (DMAF)

ITESO

jmzaratec@iteso.mx

Quantitative Finance

1 Black and Scholes Solution

Black and Scholes Solution

We defined the Black and Scholes equation for a derivative. Let's firstly Intuitively solve the fair value of a forward.

$$rf_t = \frac{\partial f_t}{\partial t} + \frac{\partial f_t}{\partial S_t} S_t r + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 f_t}{\partial S_t^2}$$

We're interesting in having the fair value in present for f_t at time T . Let's compute the expected value at T

$$f(S_T, T) := f_T = (f_T)$$

In risk neutral valuation we can compute the expected value. We can bring value to present using risk free rate.

$$f(S_t, t) := f_t = e^{-r(T-t)}(f_T)$$

Black and Scholes Solution

Let's revisit the forward case. For $f(S_t, t) := f_t = S_t$

$$rf_t = \frac{\partial f_t}{\partial t} + \frac{\partial f_t}{\partial S_t} S_t r + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 f_t}{\partial S_t^2}$$

$$rf_t = 0 + rS_t + 0$$

$$rf_t = rS_t$$

Using the value of S_t at time T

$$f_T = \mathbb{E}(f_T)$$

$$f_T = \mathbb{E}(S_T)$$

Replacing S_T . We replace μ_r by r in the risk neutral world.

$$f_T = \mathbb{E} \left[S_t e^{(\mu_r)(T-t) + \sigma \sqrt{(T-t)}(W_t)} \right]$$

Black and Scholes Solution

We replace μ by r in the risk neutral world.

$$\begin{aligned}f_T &= \mathbb{E} \left[S_t e^{(r)(T-t) + \sigma \sqrt{(T-t)}(W_n)} \right] \\f_T &= S_t e^{(r)(T-t)} \mathbb{E} \left[e^{\sigma \sqrt{(T-t)}(W_n)} \right] \\f_T &= S_t e^{(r)(T-t)} e^{(\sigma \sqrt{(T-t)}) \mathbb{E}[(W_t)]} \\f_T &= S_t e^{(r)(T-t)} e^{(\sigma \sqrt{(T-t)}) \mathbb{E}[(0)]} \\f_T &= S_t e^{(r)(T-t)}\end{aligned}$$

This is the fair value of a forward. At time t we must signed the contract of the expected value at time T , otherwise, arbitrage.

Black and Scholes Solution

We are "lucky" that our proposal of f_t satisfies the Black and Scholes equation. Let's formally solve it. There are several methods to solve differential equation for rf_t . In this class we use Feynman-Kaç theorem to solve it.

Definition: Feynman-Kaç theorem:

Problem statement:

$$\frac{\partial U}{\partial t} + \mu(x, t) \frac{\partial U}{\partial x} + \frac{1}{2} \sigma^2(x, t) \frac{\partial^2 U}{\partial x^2} - V(x, t) U(x, t) + f(x, t) = 0$$

where: $U(x, T) = \psi(x)$; therefore, $U(X_T, T) = \psi(X_T)$

Let's initialize $f(x, t) = 0$

$$rf_t = \frac{\partial f_t}{\partial t} + \frac{\partial f_t}{\partial S_t} S_t r + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 f_t}{\partial S_t^2} \quad (1)$$



ITESO
Instituto Tecnológico y de Estudios Superiores de Occidente

This is Black & Scholes differential equation for derivatives

Black and Scholes Solution

Solution:

$$U(x, t) = \mathbb{E}^Q \left[\int_t^T \left[e^{-\int_t^r V(x, \tau) d\tau} f(x_r, r) \right] dr + e^{-\int_t^T V(x, \tau) d\tau} \psi(X_T) \mid X_t : x \right]$$

Continue with this idea, let's built a solution for the differential form of f_t .

Let's propose: $U = f_t$

$$\sigma^2(x, t) = \sigma^2 S_t^2$$

$$V(x, \tau) = r$$

$$f(x_r, r) = 0$$

$$\psi(X_T) = f_T$$

$$f(x, t) = 0$$

Replacing in solution:

$$\begin{aligned}f_t &= \mathbb{E}^Q \left[\int_t^T e^{-\int_t^r(r)d\tau} (0) d_r + e^{-\int_t^T(r)d\tau} f_T \mid S_t : s \right] \\f_t &= \mathbb{E}^Q \left[e^{-\int_t^T(r)d\tau} f_T \mid S_t : s \right] \\f_t &= \mathbb{E}^Q \left[e^{-r(T-t)} f_T \mid S_t : s \right] \\f_t &= e^{-r(T-t)} \mathbb{E} [f_T]\end{aligned}$$

The expected value $\mathbb{E}_t[f_T]$ is a Valuation Neutral to Risk is the payoff $f_t(S_T, T)$ at maturity T .